



Milestone 2: Implementation, Configuration & Testing Report

Field	Detail
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Institution	North-West University (NWU)
Module	CMPG325
Client	Lesedi Renewable Energy Research Group (Mahikeng)
Client ID	CLI-109
Project ID	CMPG325-2026-109
Assigned technical challenge	Router-on-a-Stick (sub-interface inter-VLAN routing)
Deliverable	Cisco Packet Tracer (.pkt) file + report

1. Introduction

Milestone 1 produced the client requirements traceability table, the physical and logical topology designs, and the VLSM IP addressing plan for Lesedi Renewable Energy Research Group (CLI-109). This report, Milestone 2, documents the implementation of that design in Cisco Packet Tracer: configuring the switches, configuring the router's sub-interfaces for inter-VLAN routing, connecting and addressing every end device, configuring the restricted guest wireless VLAN, and testing/verifying all of it with evidence.

All configuration decisions below carry forward exactly what was approved in Milestone 1, no requirement, VLAN, or subnet was changed during implementation. Where this process surfaced practical issues (configuration slips, device state), these are documented in Section 6 (Troubleshooting) rather than hidden, since diagnosing and correcting them was itself part of the implementation work.

2. Recap of the Approved Design

2.1 VLAN & IP Addressing Plan (from Milestone 1)

VLAN	Name	Subnet	Gateway (router sub-interface)	Usable hosts
10	RESEARCH	172.30.74.0/25	172.30.74.1	126
20	ADMIN	172.30.74.128/26	172.30.74.129	62
30	SERVERS	172.30.74.192/27	172.30.74.193	30
40	GUEST-WIFI (restricted)	172.30.74.224/27	172.30.74.225	30
99	MGMT	172.30.75.0/28	172.30.75.1	14

The remainder of the 172.30.74.0/23 block (172.30.75.16 onward) is reserved for future growth and was deliberately left unallocated, as per the Milestone 1 plan.

2.2 Physical Design

- 1x Router (Router-Lesedi): single physical trunk link (FastEthernet0/0) to the Core Switch, carrying all five VLANs via 802.1Q sub-interfaces.
- 1x Core Switch: trunk aggregation point, trunk port up to the router, trunk/access ports down to the two access switches and the two VLAN 30 servers.
- 2x Access Switches: Switch A (Research wing) VLAN 10 access ports plus the uplink port for the wireless AP (VLAN 40); Switch B (Admin office) VLAN 20 access ports.
- 1x Wireless AP (AP-GUEST): connected to an existing internal port on Switch A, serving VLAN 40 only, satisfies the heritage building's no-new-external-cabling constraint (R5).
- 1x Server-Internet node: connected directly to the router on a separate link, representing the ISP/Internet boundary for testing restricted-but-not-blocked Internet access.

3. Switch Configuration

Each switch was configured with the VLANs it needed, the correct trunk encapsulation toward the Core Switch, and access ports assigned to the correct VLAN for each connected device.

3.1 Core-Switch

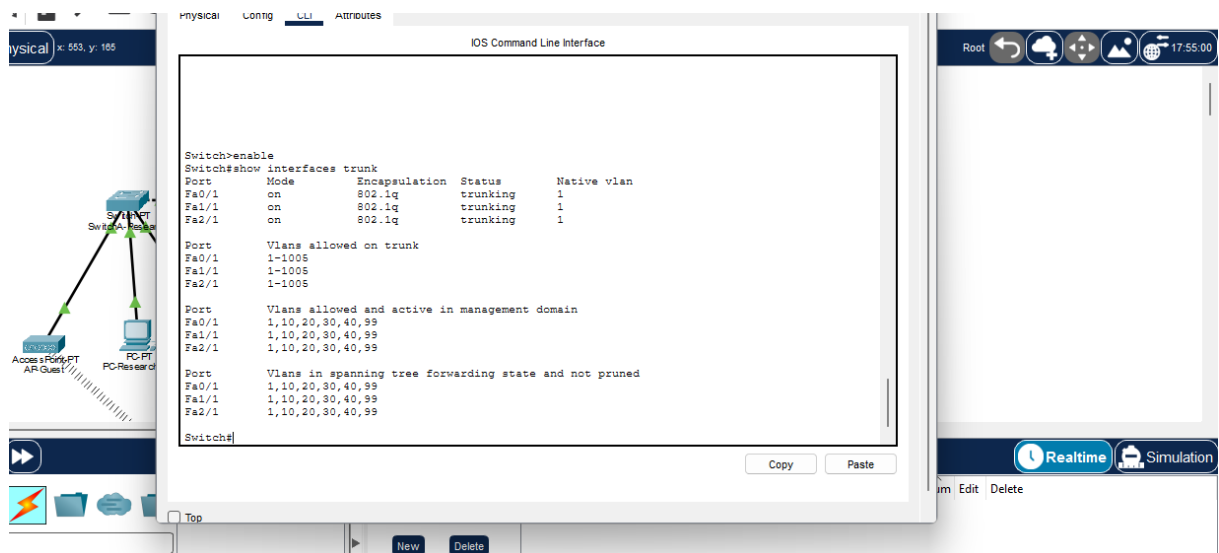
- Created VLANs 10, 20, 30, 40, 99.
- Configured the uplink to Router-Lesedi as an 802.1Q trunk carrying all five VLANs.
- Configured trunk links down to SwitchA-Research and SwitchB-Admin.
- Configured access ports (VLAN 30) for Server-File and Server-App.

3.2 SwitchA-Research

- VLAN 10 access ports for PC-Research1 and PC-Research2.
- VLAN 40 access port for the uplink to AP-GUEST (the wireless AP serving the restricted contractor VLAN).
- Trunk links up to Core-Switch.

3.3 SwitchB-Admin

- VLAN 20 access port for PC-Admin1 (confirmed via show vlan brief, VLAN 20 correctly shows Fa1/1; VLANs 10, 30, 40, 99 correctly show no ports, since none of those end devices sit on this switch).
- Trunk links up to Core-Switch.



4. Router-on-a-Stick Configuration (Router-Lesedi)

The assigned technical challenge, Router-on-a-Stick, was implemented on Router-Lesedi's FastEthernet0/0 by creating five logical sub-interfaces, one per VLAN, each with 802.1Q encapsulation matching its VLAN ID and the gateway IP address from the Milestone 1 plan.

4.1 Sub-interface configuration commands

```
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 172.30.74.1 255.255.255.128

interface FastEthernet0/0.20
encapsulation dot1Q 20
ip address 172.30.74.129 255.255.255.192

interface FastEthernet0/0.30
encapsulation dot1Q 30
ip address 172.30.74.193 255.255.255.224

interface FastEthernet0/0.40
encapsulation dot1Q 40
ip address 172.30.74.225 255.255.255.224

interface FastEthernet0/0.99
encapsulation dot1Q 99
ip address 172.30.75.1 255.255.255.240

interface FastEthernet0/0
no shutdown
```

A sixth interface, FastEthernet1/0, was additionally configured with IP 203.0.113.1/30 to connect to the Server-Internet node, representing the ISP-facing side of the network for testing the guest VLAN's permitted Internet access (see Section 5).

4.2 Verification

The configuration was saved with copy running-config startup-config, and verified with show ip interface brief, confirming all five sub-interfaces plus FastEthernet0/0 and FastEthernet1/0 showing Status/Protocol = up/up with the correct IP addresses.

The screenshot displays the Router-Lesedi interface with the CLI tab selected. The configuration commands for the sub-interfaces are shown, followed by the output of the `show access-lists 140` and `show ip interface brief` commands.

```
Router(config-if)#no shutdown
Router(config-if)#end
Router#
!SYS-S-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show access-lists 140
Extended IP access list 140
deny ip 172.30.74.224 0.0.0.31 172.30.74.0 0.0.0.127 (4 match(es))
deny ip 172.30.74.224 0.0.0.31 172.30.74.128 0.0.0.63 (4 match(es))
deny ip 172.30.74.224 0.0.0.31 172.30.74.192 0.0.0.31 (4 match(es))
deny ip 172.30.74.224 0.0.0.31 172.30.75.0 0.0.0.15
permit ip any any (4 match(es))

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset    up          up
FastEthernet0/0.10 172.30.74.1     YES manual  up          up
FastEthernet0/0.20 172.30.74.129   YES manual  up          up
FastEthernet0/0.30 172.30.74.193   YES manual  up          up
FastEthernet0/0.40 172.30.74.225   YES manual  up          up
FastEthernet0/0.99 172.30.75.1     YES manual  up          up
FastEthernet1/0 203.0.113.1     YES manual  up          up
Serial12/0      unassigned      YES unset    administratively down down
Serial13/0      unassigned      YES unset    administratively down down
FastEthernet4/0 unassigned      YES unset    administratively down down
FastEthernet5/0 unassigned      YES unset    administratively down down
Router#
```

5. Restricted Guest Wireless Access (VLAN 40)

Requirement R4 specifies that after-hours cleaning and security contractors need wireless access, but that access must be limited, not full access to research, admin, or server resources. This was implemented in two parts: a wireless AP delivering VLAN 40 connectivity, and an access control list (ACL) on the router restricting what VLAN 40 can reach.

5.1 Wireless AP configuration

- AP-GUEST configured with SSID "Lesedi-Guest", Port 1 (Radio) enabled.
- Authentication left open (disabled) at the wireless layer, since the actual access restriction is enforced at Layer 3 by the router ACL, not by Wi-Fi security, appropriate for this assignment's scope.
- Laptop-Contractor fitted with a WPC300N wireless module and connected to the "Lesedi-Guest" SSID.
- Laptop-Contractor assigned static IP 172.30.74.226 / 255.255.255.224, gateway 172.30.74.225 (VLAN 40).

5.2 Restriction ACL (Gi0/0.40 / Fa0/0.40)

An extended, numbered ACL (140) was created to permit VLAN 40 traffic out toward the Internet while denying it access to all four internal VLANs:

```
access-list 140 deny ip 172.30.74.224 0.0.0.31 172.30.74.0 0.0.0.127
access-list 140 deny ip 172.30.74.224 0.0.0.31 172.30.74.128 0.0.0.63
access-list 140 deny ip 172.30.74.224 0.0.0.31 172.30.74.192 0.0.0.31
access-list 140 deny ip 172.30.74.224 0.0.0.31 172.30.75.0 0.0.0.15
access-list 140 permit ip any any
```

```
interface FastEthernet0/0.40
ip access-group 140 in
```

ACL line	Effect
deny ... 172.30.74.0 0.0.0.127	Blocks VLAN 40 → VLAN 10 (Research, /25)
deny ... 172.30.74.128 0.0.0.63	Blocks VLAN 40 → VLAN 20 (Admin, /26)
deny ... 172.30.74.192 0.0.0.31	Blocks VLAN 40 → VLAN 30 (Servers, /27)
deny ... 172.30.75.0 0.0.0.15	Blocks VLAN 40 → VLAN 99 (Management, /28)
permit ip any any	Allows everything else, in particular, the Internet-facing link

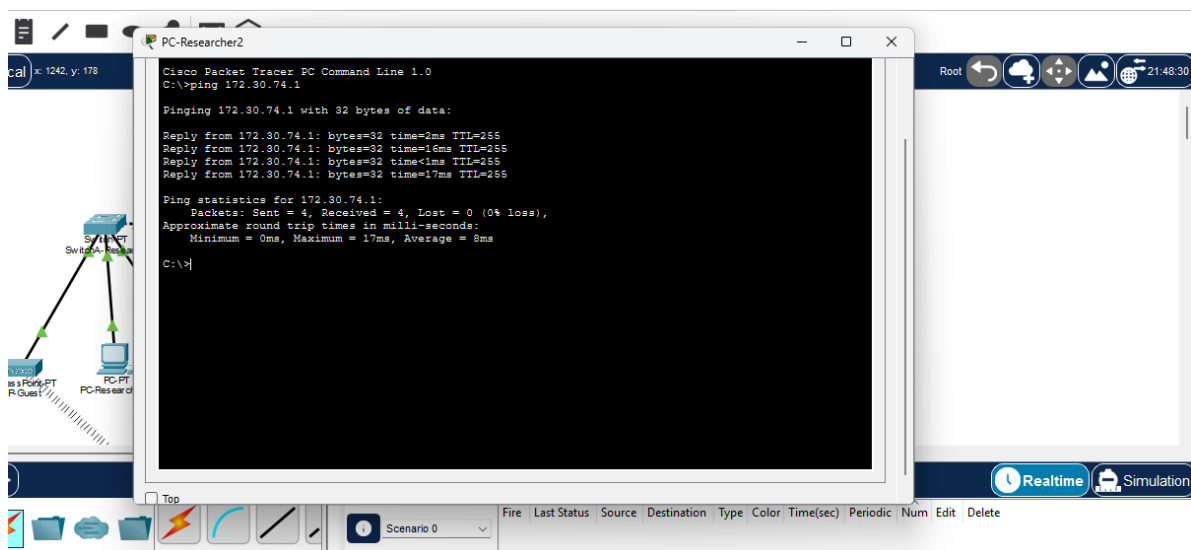
The ACL was applied inbound (in) on Fa0/0.40, filtering traffic as it arrives at the router from VLAN 40 devices, and the configuration was saved to startup-config.

6. Testing & Verification

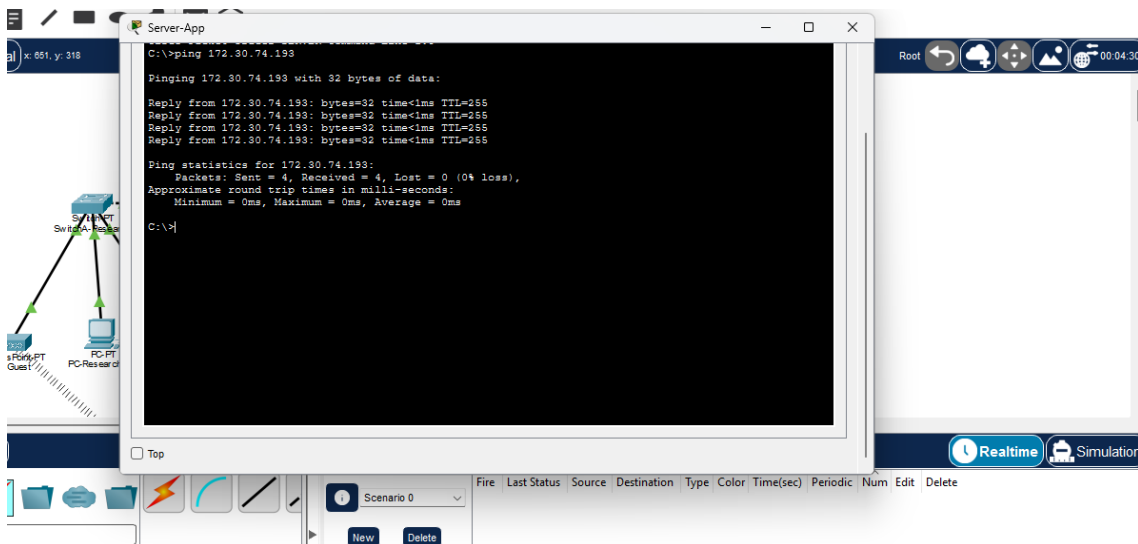
Every VLAN and the restriction policy were tested end-to-end using ping, with results captured as evidence.

6.1 Intra-VLAN connectivity (device → own gateway)

Device	VLAN	Target	Result
PC-Admin1	20	172.30.74.129	4/4 replies, 0% loss, TTL=255
PC-Research1	10	172.30.74.1	4/4 replies, 0% loss, TTL=255
PC-Research2	10	172.30.74.1	4/4 replies, 0% loss, TTL=255
Server-File	30	172.30.74.193	4/4 replies, 0% loss, TTL=255
Server-App	30	172.30.74.193	4/4 replies, 0% loss, TTL=255
Laptop-Contractor (wireless)	40	172.30.74.225	4/4 replies, 0% loss



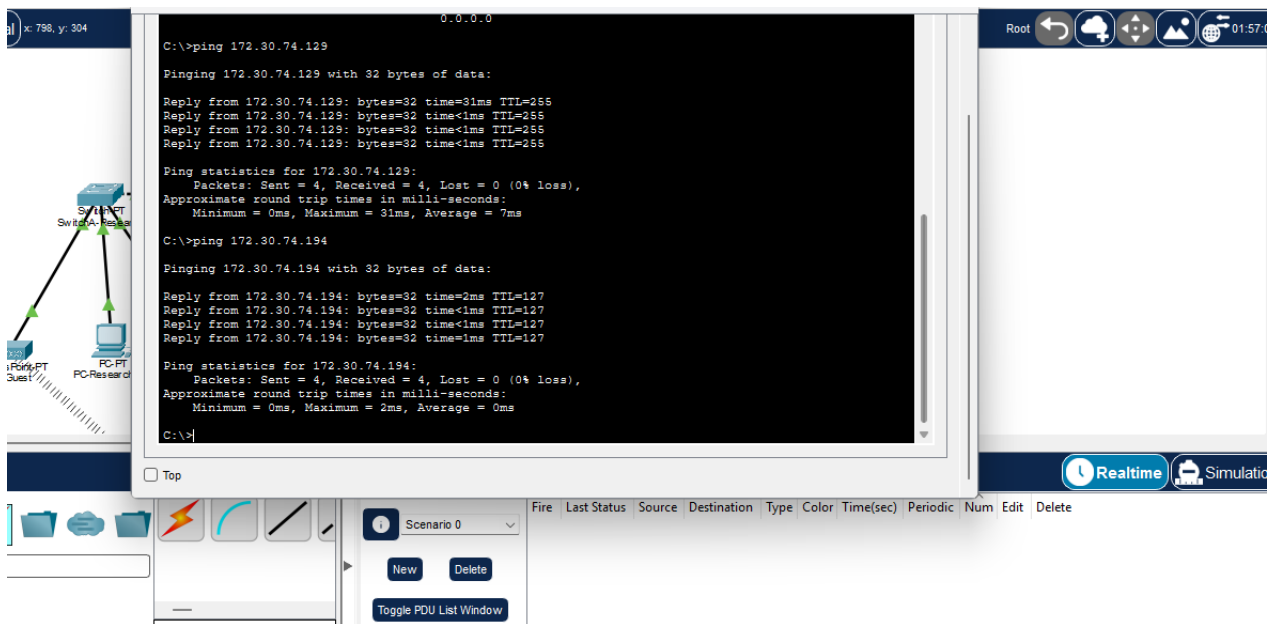
Picture 1: Ping for PC-Researcher



Picture 2: Ping for Server-App

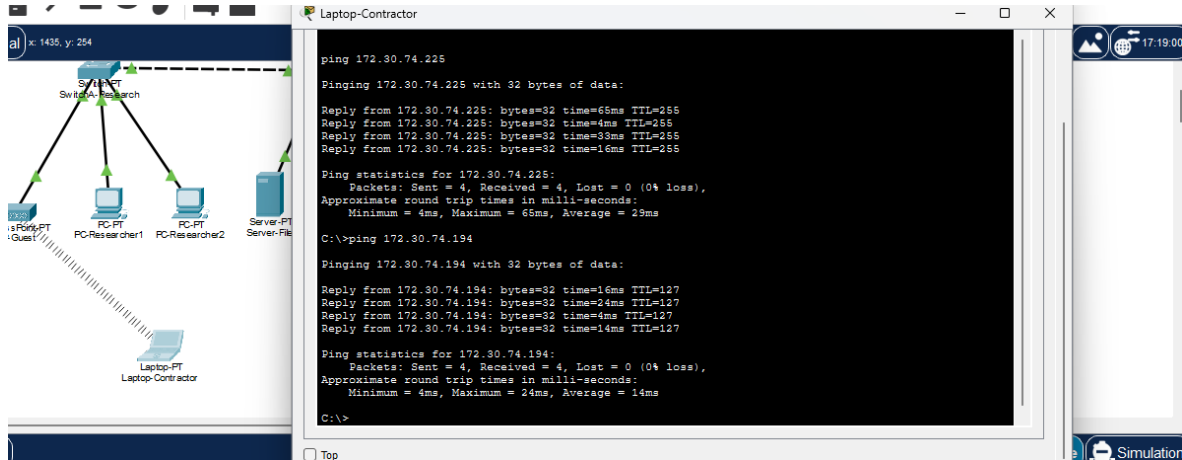
6.2 Inter-VLAN routing (Router-on-a-Stick proof)

PC-Admin1 (VLAN 20) was pinged against Server-File (VLAN 30, 172.30.74.194): 4/4 replies, 0% loss, TTL=127, the drop from the usual TTL=255 (Cisco default) to 127 (a Windows/PC default of 128 minus one hop) is direct evidence that the packet was routed through Router-Lesedi, proving the Router-on-a-Stick configuration performs real inter-VLAN routing, not simple same-subnet switching.

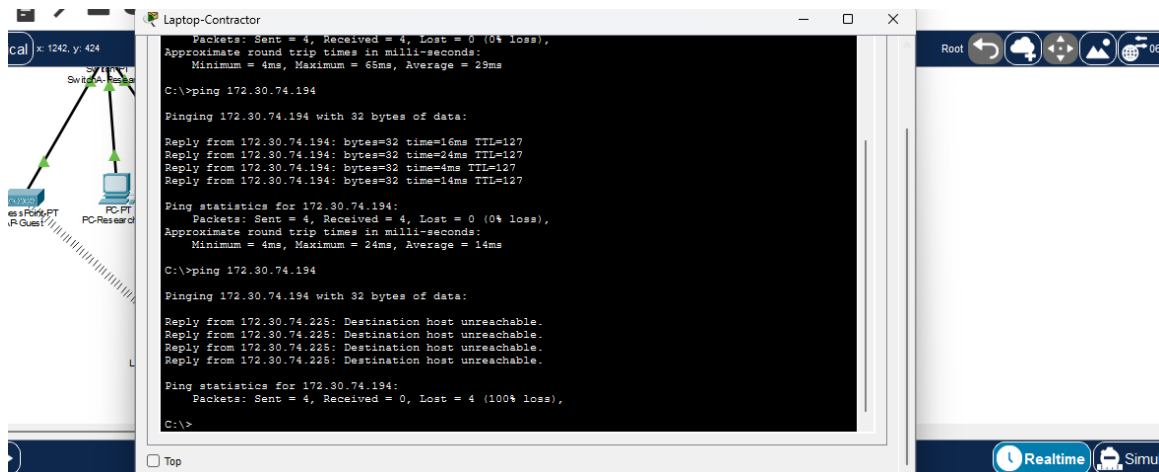


6.3 Restricted VLAN 40: before and after the ACL

Before the ACL was applied, Laptop-Contractor (VLAN 40) could reach Server-File (VLAN 30) without restriction showing 4/4 replies, TTL=127 and confirming the network was fully open prior to the restriction being configured.



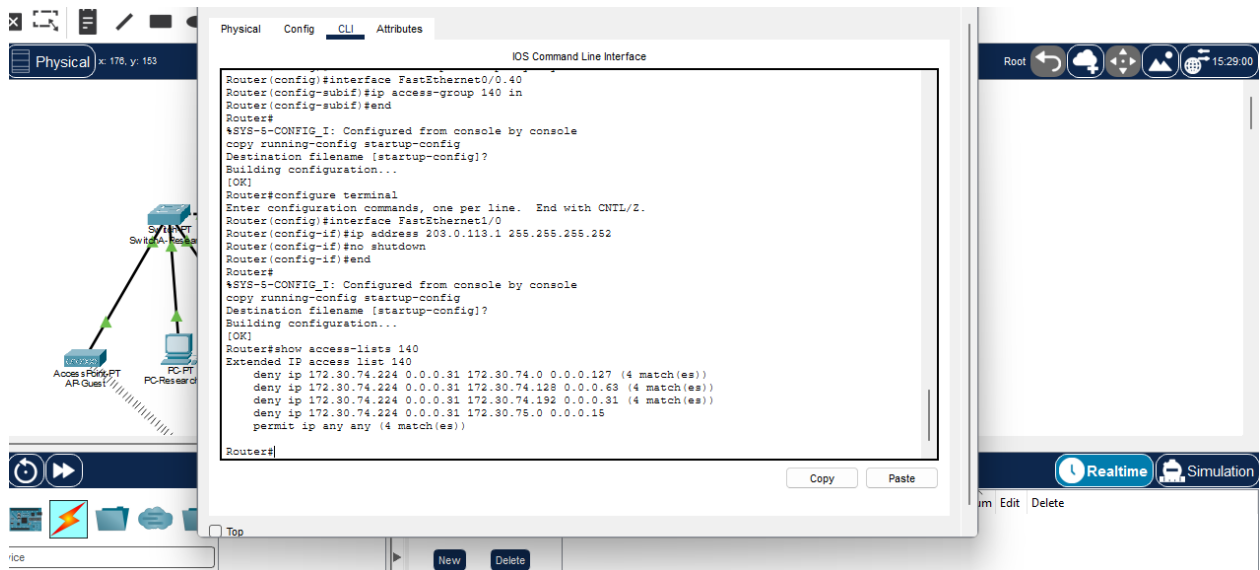
After applying ACL 140 to Fa0/0.40, the identical ping was repeated and failed, with the router itself replying "Destination host unreachable" from 172.30.74.225 (the router's own VLAN 40 gateway), proof the router is actively denying the packet due to the ACL, not simply dropping it silently. The same result was confirmed against VLAN 10 (172.30.74.2) and VLAN 20 (172.30.74.130).



Finally, Laptop-Contractor was pinged against the Server-Internet node (203.0.113.2) to confirm the guest VLAN retains permitted access: 3/4 replies succeeded (TTL=127; the single initial timeout is the normal one-off ARP resolution delay, not a fault), proving the restriction is selective, VLAN 40 is blocked from internal resources but still permitted outbound Internet access, exactly as required by R4 ("limited," not "no," access).

6.4 ACL match counters

show access-lists 140 was run after the above tests and shows match counters against each line, directly corresponding to the test pings sent 4 matches on each deny line (from the four blocked pings to VLAN 10/20/30/99) and 4 matches on the permit line (from the four pings allowed through to the Internet node). This confirms the ACL is actively filtering traffic rather than sitting unused.



7. Troubleshooting Encountered

The following real issues came up during implementation and were diagnosed and resolved; they are documented here as part of the implementation record rather than omitted.

7.1 Subnet mask truncation while typing router commands

While configuring the VLAN 20 sub-interface, the ip address command repeatedly failed with "% Invalid input detected at '^' marker." Close inspection of the exact CLI text showed the subnet mask was being entered as 255.255.192 or 255.192 instead of the full four-octet 255.255.255.192, one or two octets were being dropped while typing. The fix was to slow down and enter the command in deliberate pieces, double-checking the full four-part mask (three 255s and the final octet) before pressing Enter.

7.2 VLAN 10 sub-interface missing after a session timeout

After Packet Tracer's CLI timed out due to inactivity (dropping the session back to Router> user mode), a subsequent show ip interface brief showed four sub-interfaces (.20, .30, .40, .99) but not .10. The sub-interface was recreated with the same three commands (interface, encapsulation, ip address) and confirmed present on the next verification pass, no configuration was lost elsewhere, and the running configuration was saved immediately afterward with copy running-config startup-config to prevent a repeat.

7.3 PC/server IP addresses defaulting to 0.0.0.0

Initial ping tests from PC-Admin1 failed completely, and ipconfig showed an IPv4 address of 0.0.0.0, the device had never been given a static IP matching its VLAN. The IP Configuration utility was set to Static and the correct address/mask/gateway for each device's VLAN was entered; this was required for every end device (both PCs, both servers, and the contractor laptop).

7.4 Wireless NIC and DHCP on Laptop-Contractor

Laptop-Contractor initially had no wireless interface installed, so the AP's SSID could not be detected; a WPC300N wireless module was added via the Physical tab (with the device powered off to swap modules, then powered back on). Separately, its IP Configuration was found set to DHCP, which failed and fell back to an APIPA address (169.254.x.x); switching

the radio button to Static made the address fields editable and allowed the correct VLAN 40 address to be entered.

8. Requirements Traceability: Milestone 2

Ref	Requirement	Evidence
R1	172.30.74.0/23 addressing	All device and gateway addresses fall within the assigned block (Section 2.1, 4.1)
R2	Logical segmentation by function	Five VLANs configured and verified via show vlan brief (Section 3)
R3	Router-on-a-Stick, one sub-interface per VLAN	Five 802.1Q sub-interfaces on Fa0/0, all up/up (Section 4)
R4	Limited wireless access for contractors	ACL 140 tested: denied to VLAN 10/20/30/99, permitted to Internet (Section 5–6)
R5	No new external-wall cabling	AP-GUEST connected via existing internal port on SwitchA-Research
R6	Fully buildable/testable in Packet Tracer	CMPG325-CLI-109-Lesedi-Network.pkt — this entire report built and tested in Packet Tracer
R7	End-to-end connectivity demonstrated	Intra-VLAN, inter-VLAN (TTL=127), and restricted-VLAN before/after tests (Section 6)

9. Conclusion

All design decisions from Milestone 1 were implemented without deviation and verified with direct evidence: every VLAN's end devices can reach their own gateway, inter-VLAN routing through Router-Lesedi's Router-on-a-Stick configuration is proven via the TTL drop to 127, and the contractor's restricted wireless VLAN is proven to be blocked from internal resources while still permitted outbound Internet access. The network as built satisfies every traced client requirement (R1–R7).

Deliverables accompanying this report: CMPG325-CLI-109-Lesedi-Network.pkt (the working Packet Tracer file) and the evidence screenshots referenced throughout this document.